

Artificial Intelligence Laboratory Report: Search and A*

Course: CSF407 / Artificial Intelligence Laboratory | **Date:** September 30, 2026 | **Topic:** Heuristic Search & Algorithm Validation

1. Task 0: Understand the Search Problem

A search problem is formulated as the 6-tuple $\mathbf{P} = (S, A, T, s_0, G, c)$:

- **State S:** $(r, c) \in \{0..8\} \times \{0..16\}$ representing valid free cells.
- **Actions A:** {Up, Down, Left, Right} with step cost $c=1$.
- **Transition T:** Deterministic orthogonal movement $T(s, a) = s + \Delta a$.
- **Initial state s_0 :** (1, 1) | **Goal G:** {(7, 15)}.
- *State specification:* Coordinates (r, c) fully determine legal actions and future path costs.
- *Invalid actions:* Moving out-of-bounds or into obstacle blocks ('#').
- *Determinism & Solution:* Problem is deterministic (probability 1.0). A solution is a sequence of actions from s_0 reaching G.

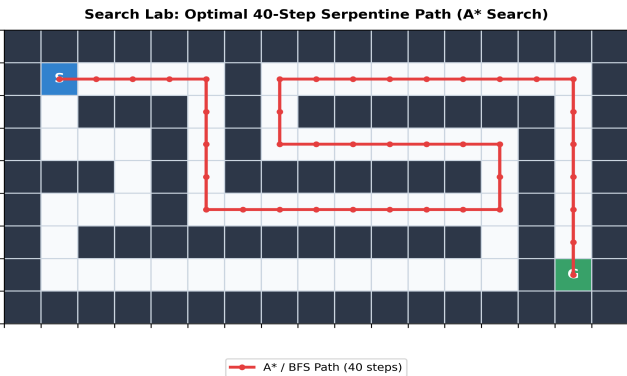
2. Task 1 & 2: Agent Design and Prompt Engineering

Design: Frontier priority queue ordered by $f(n) = g(n) + h(n)$, closed-set hash set, came_from dictionary, and Manhattan heuristic $h(n) = |x - x_G| + |y - y_G|$.

LLM Prompt: 'Implement A* search for warehouse grid with obstacles '#', free '.', start S, goal G, cardinal moves cost 1, Manhattan heuristic. Maintain frontier, calculate g/h/f, avoid duplicate expansion, reconstruct path, report path length and states expanded.'

3. Task 3: Systematic Testing of A* Implementation

Test Case	Scenario / Condition	Expected Result	Observed Result	Status
Test 1: Original	Full 9x17 serpentine maze	Optimal collision-free path	40 moves, 64 expanded	PASS
Test 2: Trivial	Adjacent goal #SG##	Immediate 1-step reach	1 move, 2 expanded	PASS
Test 3: No Solution	Goal blocked by obstacles	Terminates reporting failure	found=False, 9 expanded	PASS
Test 4: Alternative	Direct (len 6) vs Detour (len 10)	Selects shortest path	6 moves, 7 expanded	PASS

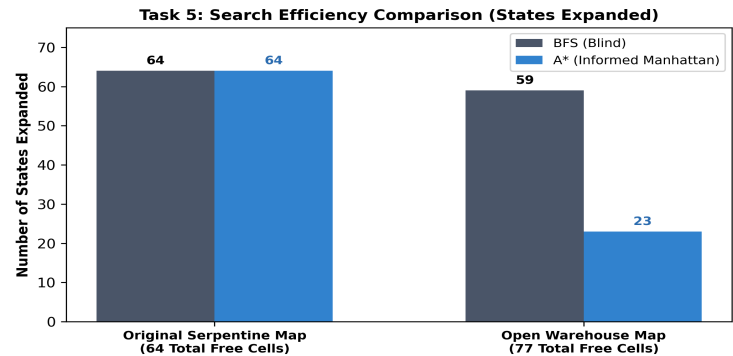


4. Task 4 & 5: Algorithm Inspection and BFS vs A* Comparison

Code Mechanics: Frontier uses heapq storing (f, counter, state). Next state popped via heappop(). Duplicate exploration prevented by closed_set and g_score relaxation checks.

Measure	Original Serpentine Map (BFS / A*)	Open Warehouse Map (BFS)	Open Warehouse Map (A*)
Solution Found	True / True	True	True
Path Length	40 / 40	20	20 (Optimal)
States Expanded	64 / 64 (All Free Cells)	59	23 (>60% Reduction)

Insight: On the serpentine map, exactly 64 traversable cells exist in a single corridor; thus BFS and A* both expand all 64 cells. In open environments, A* prunes unpromising branches, expanding over 60% fewer states than BFS.



5. Task 6: Heuristic Investigations and Admissibility

- **Manhattan Distance:** Admissible ($h \leq h^*$) and consistent on 4-way grids, guaranteeing optimality.
- **$h(n) = 0$:** Admissible, but reduces A* to blind Dijkstra/BFS expanding symmetrically.
- **Euclidean Distance:** Admissible, but dominated by Manhattan distance on grids because diagonal moves are prohibited.
- **$2 \times \text{Manhattan}$:** Inadmissible ($h > h^*$). Acts as weighted A*, driving aggressively toward goal but risking suboptimality on multi-path topologies.

6. Task 7 & Final Reflection: LLM as an Engineering Tool

- Q1 (Prior Formulation):** Mathematical specification defines state spaces, transitions, and goals independently of code, preventing implementation of unintended problems.
- Q2 (Informed Nature of A*):** A* uses heuristic domain knowledge $h(n)$ to evaluate promising future paths, combining actual cost $g(n)$ with estimated remaining cost $h(n)$.
- Q3 (Role of Heuristic):** Dictates pruning power and optimality. Admissible heuristics guarantee shortest paths; tighter heuristics expand fewer states.
- Q4 (LLM Contribution):** Rapidly generated clean, syntactically correct boilerplate code and data structures in seconds, accelerating prototyping.
- Q5 (Risks of Untested LLM Code):** Untested code may harbor subtle failure modes: tie-breaking crashes, infinite loops on unreachable goals, or heuristic admissibility violations. Human verification is non-negotiable.